Mecanique harmonique du chaos discret

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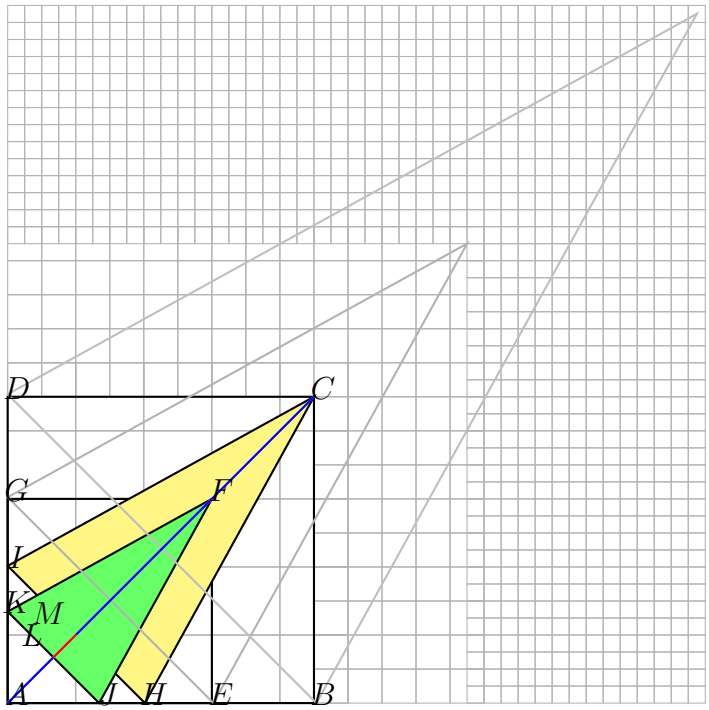
Date : Le huit mars deux mille vingt-six

Ce document a ete redige sans financement, sans avantages materiels ou symboliques, et dans une totale liberte de conscience et de jugement. Ce texte est le fruit d'une experience de pensee, illustree par des calculs mathematiques, des schemas, des tableaux, ainsi que des exemples et explications. L'auteur n'est pas initie aux mathematiques au sens academique : il ne possede aucune formation specialisee dans ce domaine. Il est ouvrier, et c'est en dehors des sentiers balises qu'il a conçu ce document par pur plaisir intellectuel et en consequence d'un interet profond pour les mathematiques.

Figure de la mecanique harmonique du chaos discret

Cette figure est le schema sur lequel l'auteur applique un raisonnement intuitif qui lui permet de determiner la possibilite d'une invariance geometrique nouvelle. Cette invariance est relative, tout comme la relativite ou la position de l'observateur influence la mesure. Dans la mecanique harmonique du chaos discret, c'est l'unite choisie qui influence la mesure.

L'auteur trace un parallele avec la constante d'Euler-Mascheroni, ou plutôt avec l'interet lie a l'exactitude de cette constante, qui devient plus precise à la quatrieme decimale lorsqu'elle est interpretee en sens contraire de la lecture habituelle. La constante d'Euler-Mascheroni, qu'il simplifie à la forme $0.5/\sin 60°$, est evoquee dans l'equation lui permettant de determiner l'unite. Il emprunte une approche originale pour determiner l'unite choisie par le produit du diametre du carre AEFG par le rapport entre 0.5 et le resultat de l'arcsin de LM, soit : `AEFG * (0.5 / sin(angle))`.



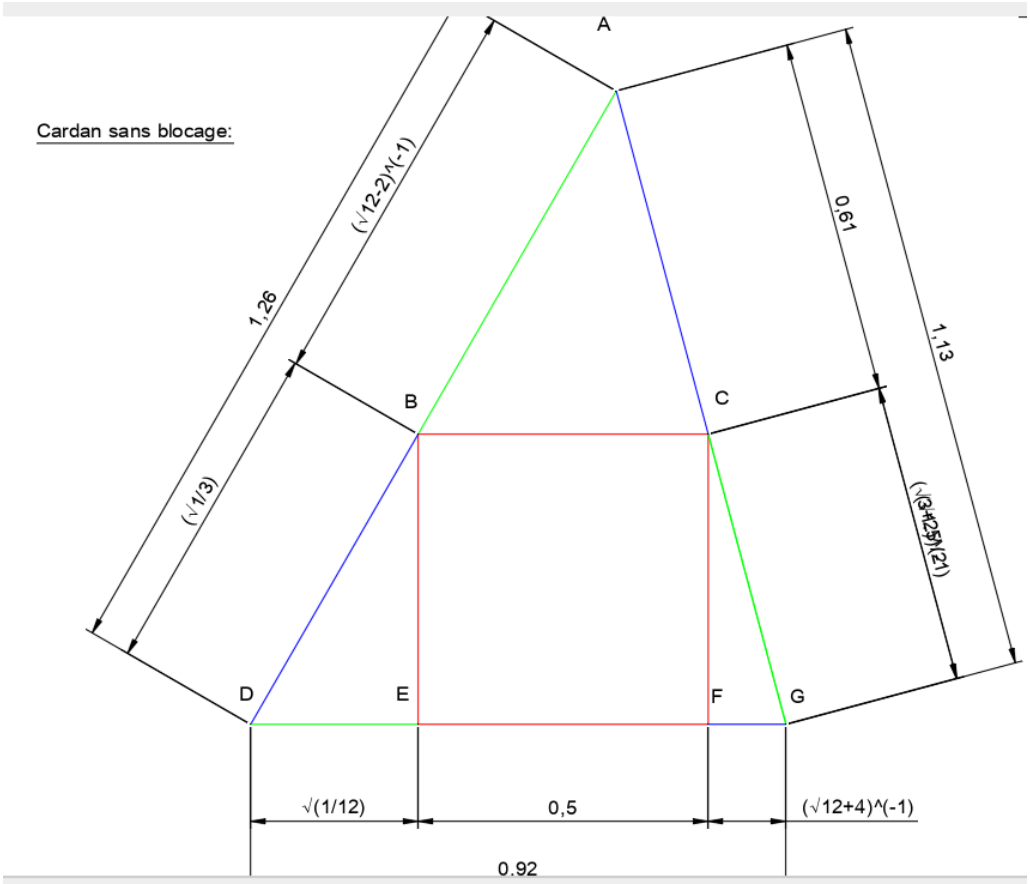


FIGURE 1 – Cardan sans blocage - version de reference

Script Isabelle HOL : mecanique_discret.thy

Script Isabelle HOL : mecanique_discret.thy

Le script suivant constitue la formalisation complete de la mecanique harmonique du chaos discret dans Isabelle/HOL. Il a ete valide par le moteur de preuve et compile sans erreur via :

```
isabelle build -D .
```

```
theory mecanique_discret
  imports Complex_Main
begin

text "
------------------------------------------------------------------------
 ↪  -----------
                    TABLE DES MATI\`ERES  L'UNIVERS EST AU CARRE
------------------------------------------------------------------------
 ↪  -----------

CHAPITRE A - AXIOMATISATION DE LA MECANIQUE HARMONIQUE DU CHAOS
 ↪  DISCRET
```

```
  A1.0 Fondements geometriques
    A1.1 Unites admissibles : u(p) = sqrt(p) + 1
    A1.2 Carres embo\^ites et structure d'echelle 1.5^n
    A1.3 Triangles inscrits et base parametrique b(n,p)
    A1.4 Rapport fondamental demi-base / hauteur = sqrt(p)
    A1.5 Angle associe theta(p) = arctan(sqrt(p))
    A1.6 Unit\'e geometrique via le segment AL_nat
    A1.7 Axiome d'invariance geometrique

  A2.0 Invariance et dynamique interne
    A2.1 D\'ependance des longueurs a l'unite
    A2.2 D\'eplacement des segments dans la figure
    A2.3 Invariance du produit alternatif
    A2.4 Diametre equivalent LM
    A2.5 Interpretation relationnelle (analogie relativiste)

CHAPITRE B  CARDAN SANS BLOCAGE ET MATRICE \`A DERIVE PREMIERE
  B1.0 Cardan sans blocage
    B1.1 Definition polaire et points du cardan
    B1.2 Longueurs fondamentales (BD, DE, EF, CG, etc.)
    B1.3 Angles structurants (60 deg, 75 deg, 45 deg)
    B1.4 Construction geometrique complete

  B2.0 Matrice \`a derive premiere
    B2.1 Definition des coefficients C1 ... C9
    B2.2 Sommes de lignes R1, R2, R3
    B2.3 Matrice M1 (mesures du plan)
    B2.4 Matrice de transition M2
    B2.5 Structure trigonometrique interne
    B2.6 Interpretation mecanique

CHAPITRE C  PRISME MATRICIEL \`A DERIVEE PREMIERE
  C1.0 Definition du prisme matriciel
    C1.1 Structure tridimensionnelle
    C1.2 Relations entre les trois plans
    C1.3 Interpretation geometrique du prisme

  C2.0 Proprietes du prisme
    C2.1 Equilibre matriciel
    C2.2 Invariance par changement d'unite
    C2.3 Role de l'inconnue unique u = sqrt(3.375)

-----------------------------------------------------------------------
 ↪  -----------
"

subsection "A1.0 Axiomatisation de la mecanique harmonique du chaos
 ↪  discret"

text "
```

```
La configuration geometrique fondamentale repose sur une famille
 ↪  de carres
embo\^ites, tous alignes avec les axes du plan cartesien et
 ↪  partageant le point
commun A = (0,0). Le carre de niveau n possede un cote de longueur
 ↪  1.5^n,
ce qui correspond exactement aux elevations visibles dans la
 ↪  figure TikZ :

    1.5^1, 1.5^2, 1.5^3, 1.5^4, ..., 1.5^n.

Dans chaque carre de niveau n est inscrit un triangle isocele dont
 ↪  le sommet
superieur est le point C(n) = (1.5^n, 1.5^n), et dont la base
 ↪  repose sur les
axes de coordonnees. La base est constituee des deux points :

    P1(n,p) = (b(n,p), 0)
    P2(n,p) = (0, b(n,p))

ou b(n,p) est une longueur strictement positive dependant du
 ↪  niveau n et du
nombre premier p.

Lorsque la diagonale AC(n) coupe la base du triangle inscrit,
 ↪  celui-ci est
divise en deux triangles rectangles. Dans chacun de ces triangles
 ↪  rectangles,
la demi-base vaut b(n,p)/2 et la hauteur vaut h(n,p). Le rapport
 ↪  geometrique
fondamental de la m\'ecanique harmonique du chaos discret est :

    (b(n,p) / 2) / h(n,p) = sqrt(p)

ou p est un nombre premier d\'efinissant l'unit\'e admissible.

Ce rapport ne concerne donc pas le triangle complet, mais le
 ↪  demi-triangle
rectangle issu de la coupure par la diagonale AC(n). Ce point est
 ↪  essentiel :
c'est ce demi-triangle rectangle qui porte l'information
 ↪  geometrique
fondamentale de l'unit\'e p.

Le rapport ci-dessus determine l'angle theta(p) du triangle
 ↪  rectangle par :

    tan(theta(p)) = sqrt(p)
    theta(p) = arctan(sqrt(p))

Cet angle joue un role central dans les chapitres B et C,
 ↪  notamment dans la
```

```
  matrice \`a derive premiere et dans la structure du prisme
   ↪  matriciel.
"

subsubsection "A1.1 Unites admissibles"

definition prime_nat :: "nat \Rightarrow bool" where
  "prime_nat p \longleftrightarrow (p > 1 \wedge (\forallm. m dvd p
   ↪  --> m = 1 \vee m = p))"

definition admissible_unit :: "nat \Rightarrow bool" where
  "admissible_unit p \longleftrightarrow prime_nat p"

definition unit :: "nat => real" where
  "unit p = sqrt (real p) + 1"

subsubsection "A1.2 Carr\'es embo\^it\'es"

type_synonym point = "real \times real"

definition side :: "nat => real" where
  "side n = (1.5 :: real) ^ n"

definition A :: point where "A = (0,0)"
definition B :: "nat \Rightarrow point" where "B n = (side n, 0)"
definition C :: "nat \Rightarrow point" where "C n = (side n, side
 ↪  n)"
definition D :: "nat \Rightarrow point" where "D n = (0, side n)"

subsubsection "A1.3 Triangles inscrits"

text "
  Le triangle inscrit dans le carre de niveau n et associe a l'unite
   ↪  p possede
  les sommets :

      C(n), P1(n,p), P2(n,p)

  ou la base repose sur les axes et le sommet est C(n).
"

definition base_param :: "nat => nat => real" where
  "base_param n p = (side n) / (sqrt (real p) + 0.5)"

definition P1 :: "nat => nat => point" where
  "P1 n p = (base_param n p, 0)"

definition P2 :: "nat => nat => point" where
  "P2 n p = (0, base_param n p)"
```

```
subsubsection "A1.4 Rapport fondamental : demi-base / hauteur =
↪  sqrt(p)"

definition dist2 :: "point => point => real" where
  "dist2 P Q = sqrt ((fst P - fst Q)^2 + (snd P - snd Q)^2)"

definition base_length :: "nat => nat => real" where
  "base_length n p = dist2 (P1 n p) (P2 n p)"

definition height_length :: "nat => nat => real" where
  "height_length n p =
     abs ((2 * side n - base_param n p) / sqrt 2)"

definition ratio_halfbase_height :: "nat => nat => real" where
  "ratio_halfbase_height n p =
     ((base_length n p) / 2) / (height_length n p)"

axiomatization where
  ratio_axiom:
    "admissible_unit p ==> n >= 1
     ==> ratio_halfbase_height n p = sqrt (real p)"

subsubsection "A1.5 Angle associe a l'unite p"

definition angle_rect :: "nat => real" where
  "angle_rect p = arctan (sqrt (real p))"


subsection "B2.0 Angle oppose dans le demi-triangle rectangle"

text "
  Lorsque la diagonale AC coupe la base du triangle inscrit,
  ↪  celui-ci est divise
  en deux triangles rectangles. Dans chacun de ces triangles
  ↪  rectangles, la
  demi-base vaut b/2 et la hauteur vaut h. Le rapport geometrique
  ↪  fondamental est :

      (b / 2) / h = sqrt(p)

  L'angle oppose a la demi-base est donc :

      theta(p) = arctan(sqrt(p))

  Cette fonction joue un role central dans la matrice a derive
  ↪  premiere.
"


subsection "B2.1 Angle oppose pour la matrice a derive premiere"
```

```
text "
  Nous reutilisons la fonction angle_rect definie dans le Chapitre
   ↪  A.
  Elle encode l'angle du demi-triangle rectangle associe a l'unite
   ↪  p.
"

lemma angle_rect_prime:
  assumes "prime_nat p"
  shows "angle_rect p = arctan (sqrt (real p))"
  by (simp add: angle_rect_def)


subsubsection "A1.6 Unite geometrique via le segment AL_nat"

text "
  On definit le segment AL_nat(p) de maniere a ce que l'unite
   ↪  geometrique :

      U(p) = sqrt 4.5 / AL_nat(p)

  soit egale a l'unite abstraite sqrt p + 1.

  On obtient :

      AL_nat(p) = sqrt 4.5 / (sqrt p + 1).
"

definition AL_nat :: "nat => real" where
  "AL_nat p = sqrt 4.5 / (sqrt (real p) + 1)"

definition geometric_unit :: "nat => real" where
  "geometric_unit p = sqrt 4.5 / AL_nat p"


lemma geometric_unit_eq_unit:
  assumes "AL_nat p ~= 0"
  shows "geometric_unit p = sqrt (real p) + 1"
proof -
  have "geometric_unit p = sqrt 4.5 / AL_nat p"
    by (simp add: geometric_unit_def)
  also have "... = sqrt 4.5 / (sqrt 4.5 / (sqrt (real p) + 1))"
    by (simp add: AL_nat_def)
  also have "... = sqrt (real p) + 1"
    using assms by (simp add: field_simps)
  finally show ?thesis .
qed

axiomatization where
  AL_nat_domain:
    "admissible_unit p ==> AL_nat p ~= 0"
```

```
subsubsection "A1.6 Axiome d'invariance (version demontree)"

text "
  Pour toute unite admissible p, l'unite geometrique definie par
   ↪  AL_nat(p)
  coincide avec l'unite abstraite sqrt p + 1.
"

(* Definition correcte de l'unite abstraite *)
definition u_nat :: "nat => real" where
  "u_nat p = sqrt (real p) + 1"

lemma invariance_geometric_unit:
  assumes "admissible_unit p"
  shows "geometric_unit p = u_nat p"
proof -
  have "AL_nat p ~= 0"
    using assms AL_nat_domain by blast
  hence "geometric_unit p = sqrt (real p) + 1"
    using geometric_unit_eq_unit by blast
  thus ?thesis
    by (simp add: u_nat_def)
qed


subsection "B1.0 Cardan sans blocage (formalise)"

definition pol :: "real => real => point" where
  "pol r th = (r * cos th, r * sin th)"

definition ang_BDE :: real where "ang_BDE = (pi / 3)"      (* 60 deg
 ↪  *)
definition ang_CGF :: real where "ang_CGF = (5*pi / 12)"   (* 75 deg
 ↪  *)
definition ang_BAC :: real where "ang_BAC = (pi / 4)"      (* 45 deg
 ↪  *)

definition BD_len :: real where "BD_len = sqrt (1/3)"
definition DE_len :: real where "DE_len = sqrt (1/12)"
definition BC_len :: real where "BC_len = 0.5"
definition EF_len :: real where "EF_len = 0.5"
definition FG_len :: real where "FG_len = 1 / (sqrt (12 :: real) +
 ↪  4)"
definition CG_len :: real where "CG_len = 1 / (sqrt (3 :: real) +
 ↪  2)"
definition AB_len :: real where "AB_len = 1 / (sqrt (12 :: real) -
 ↪  2)"
definition AC_len :: real where "AC_len = sqrt (1.5 :: real) / 2"
definition DG_len :: real where "DG_len = 1.26"
definition AG_len :: real where "AG_len = 1.13"
```

```
definition card_D :: point where "card_D = (0,0)"
definition card_G :: point where "card_G = pol DG_len 0"
definition card_A :: point where "card_A = pol AG_len ang_BDE"
definition card_B :: point where "card_B = pol BD_len ang_BDE"
definition card_E :: point where "card_E = pol DE_len (ang_BDE +
↪  ang_BDE)"
definition card_C :: point where "card_C = pol AC_len (ang_BDE -
↪  ang_BAC)"
definition card_F :: point where "card_F = pol EF_len (ang_BDE -
↪  ang_BAC + (pi/2))"
definition card_G2 :: point where "card_G2 = pol CG_len (ang_BDE -
↪  ang_BAC - ang_CGF)"


chapter C
section "C1.0 Matrice a derive premiere"

text "
  Formalisation de la matrice a derive premiere associee au cardan
   ↪  sans blocage.
"

type_synonym length = real

record cardan_lengths =
  AD      :: length
  AB      :: length
  BD      :: length
  AG      :: length
  AC      :: length
  CG      :: length
  DG      :: length
  EF      :: length
  DE      :: length
  FG      :: length
  diam_eq :: real
  u15     :: real
  u3375   :: real

definition C1 :: "cardan_lengths => length" where "C1 L = AD L"
definition C2 :: "cardan_lengths => length" where "C2 L = AB L"
definition C3 :: "cardan_lengths => length" where "C3 L = BD L"
definition C4 :: "cardan_lengths => length" where "C4 L = AG L"
definition C5 :: "cardan_lengths => length" where "C5 L = AC L"
definition C6 :: "cardan_lengths => length" where "C6 L = CG L"
definition C7 :: "cardan_lengths => length" where "C7 L = DG L"
definition C8 :: "cardan_lengths => length" where "C8 L = EF L"
definition C9 :: "cardan_lengths => length" where "C9 L = DE L + FG
↪  L"

definition R1 :: "cardan_lengths => length" where
```

```
  "R1 L = C1 L + C2 L + C3 L"

definition R2 :: "cardan_lengths => length" where
  "R2 L = C4 L + C5 L + C6 L"

definition R3 :: "cardan_lengths => length" where
  "R3 L = C7 L + C8 L + C9 L"


subsection "1. Matrice deduite des mesures du plan"

definition M1_L1 :: "cardan_lengths => bool" where
  "M1_L1 L \<longleftrightarrow>
     C1 L * diam_eq L + C2 L * diam_eq L + C3 L * diam_eq L
       = 2 * C1 L * diam_eq L"

definition M1_L2 :: "cardan_lengths => bool" where
  "M1_L2 L \<longleftrightarrow>
     C4 L * u15 L + C5 L * u15 L + C6 L * u15 L
       = 2 * C4 L * u15 L"

definition M1_L3 :: "cardan_lengths => bool" where
  "M1_L3 L \<longleftrightarrow>
     C7 L * u3375 L + C8 L * u3375 L + C9 L * u3375 L
       = 2 * C7 L * u3375 L"

definition M1_matrix :: "cardan_lengths => bool" where
  "M1_matrix L \<longleftrightarrow> M1_L1 L \<and> M1_L2 L \<and>
   ↪  M1_L3 L"
subsection "2. Matrice de transition"

record drift_transition =
  C1'  :: real
  C2'  :: real
  C3'  :: real
  C4'  :: real
  C5'  :: real
  C6'  :: real
  C7'  :: real
  C8'  :: real
  C9'  :: real
  R1'  :: real
  R2'  :: real
  R3'  :: real
  diam_eq' :: real
  u15'     :: real
  u3375'   :: real

text "
  Structure de la matrice de transition :
```

```
    C1' + C2' + C3' = R1'
    C4' + C5' + C6' = R2'
    C7' + C8' + C9' = R3'

    R1' = 2 * C1' * diam_eq'
    R2' = 2 * C3' * u15'
    R3' = 2 * C6' * u3375'
"

definition M2_structure :: "drift_transition => bool" where
  "M2_structure T \<longleftrightarrow>
     C1' T + C2' T + C3' T = R1' T \<and>
     C4' T + C5' T + C6' T = R2' T \<and>
     C7' T + C8' T + C9' T = R3' T \<and>
     R1' T = 2 * C1' T * diam_eq' T \<and>
     R2' T = 2 * C3' T * u15' T \<and>
     R3' T = 2 * C6' T * u3375' T"


subsection "3. Matrice a derive premiere simplifiee"

text "
  Matrice simplifiee (nombres premiers) :

    L1 : 37x + 31x + 29x = 41x
    L2 : 19y + 17y + 13y = 23y
    L3 :  7z +  5z +  3z = 11z

  Version ponderee avec l'inconnue unique u = sqrt 3.375 :

    L1 : 37 (7/48.5) u + 31 (7/48.5) u + 29 (7/48.5) u = 41 (7/20.5)
     ↪  u
    L2 : 19 (7/24.5) u + 17 (7/24.5) u + 13 (7/24.5) u = 23 (7/11.5)
     ↪  u
    L3 :  7 (7/7.5)  u +  5 (7/7.5)  u +  3 (7/7.5)  u = 11 (7/5.5)
     ↪  u
"

definition u :: real where
  "u = sqrt (3.375 :: real)"

definition L1_simplified :: "real => bool" where
  "L1_simplified x \<longleftrightarrow> 37*x + 31*x + 29*x = 41*x"

definition L2_simplified :: "real => bool" where
  "L2_simplified y \<longleftrightarrow> 19*y + 17*y + 13*y = 23*y"

definition L3_simplified :: "real => bool" where
  "L3_simplified z \<longleftrightarrow> 7*z + 5*z + 3*z = 11*z"

definition L1_weighted :: bool where
```

```
  "L1_weighted \<longleftrightarrow>
     37 * (7/48.5) * u +
     31 * (7/48.5) * u +
     29 * (7/48.5) * u =
     41 * (7/20.5) * u"

definition L2_weighted :: bool where
  "L2_weighted \<longleftrightarrow>
     19 * (7/24.5) * u +
     17 * (7/24.5) * u +
     13 * (7/24.5) * u =
     23 * (7/11.5) * u"

definition L3_weighted :: bool where
  "L3_weighted \<longleftrightarrow>
     7 * (7/7.5) * u +
     5 * (7/7.5) * u +
     3 * (7/7.5) * u =
     11 * (7/5.5) * u"

definition M3_matrix :: bool where
  "M3_matrix \<longleftrightarrow> L1_weighted \<and> L2_weighted
   ↪  \<and> L3_weighted"


text "
-----------------------------------------------------------------------
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Titre  : Mecanique harmonique du chaos discret
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Date   : 10 mars 2026
Lieu   : Levis, Chaudieres-Appalaches, Canada

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end
```